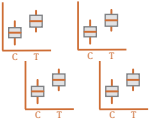
N_{study} $N_{species}$ $N_{effect\ size}$



$$y_i = \beta_0 + u_{j[i]} + p_{k[i]} + q_{k[i]} + s_i + e_i$$

$p_k \sim N(0, \sigma_p^2 A)$

$q_k \sim N(0, \sigma_q^2 I)$

$S_k \sim N(0, \sigma_s^2 I)$

$e_i \sim N(0, v_i)$

$j \approx k$

Non-identifiable...

$k \approx i$ $i \approx j$